Chapter 4 Route Map: After One Line Is Not Enough

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4 overview (KP01–KP05)

Slides: read and download

Learning objectives

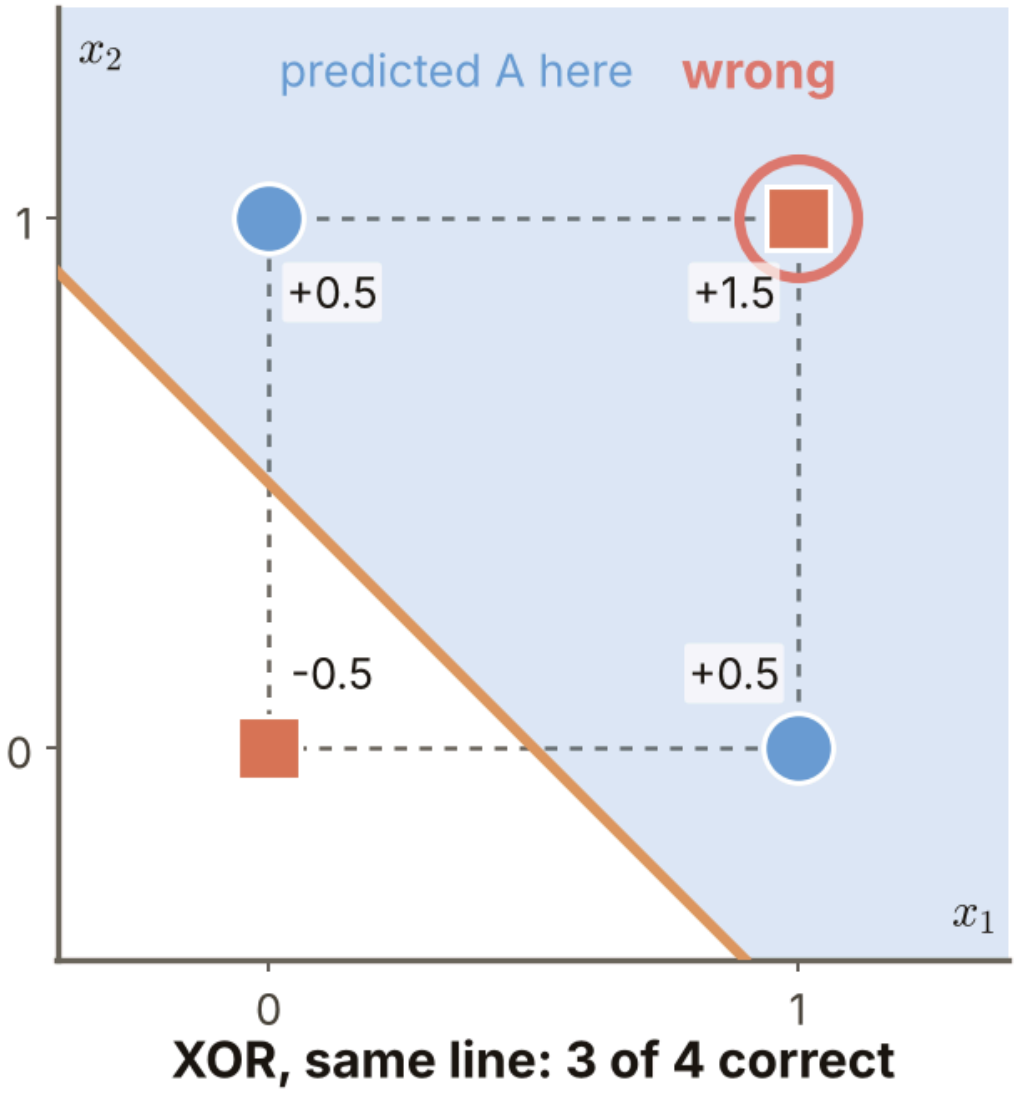
- Connect KP01 to KP05 with one question and state the question, the answer and what is left open for each.
- Say in which knowledge point each term of the chapter appears.
- State the conclusion and limits of the chapter and what the next chapter learns.

Chapter 4 Route Map

One-sentence summary: When labels alternate (as in XOR) one line fails; adding lines and layers separates them, KP05 says how a network grows from points, and the open question is how to learn the weights.

1. This chapter asks one thing

How can a network separate labels that one line cannot? The starting point is XOR: A on one diagonal, B on the other.



2. Route map: each knowledge point takes over the question left by the previous one

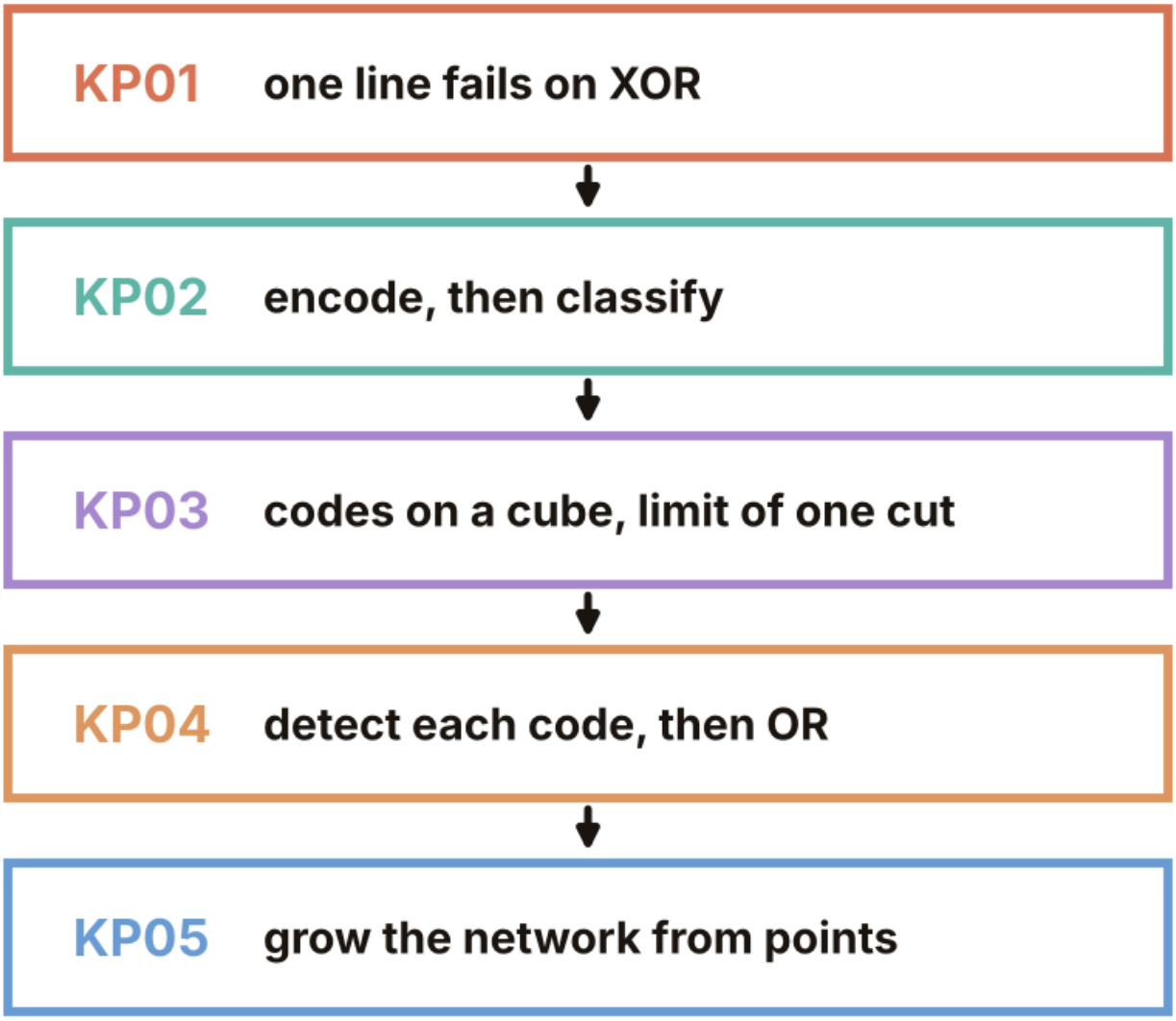
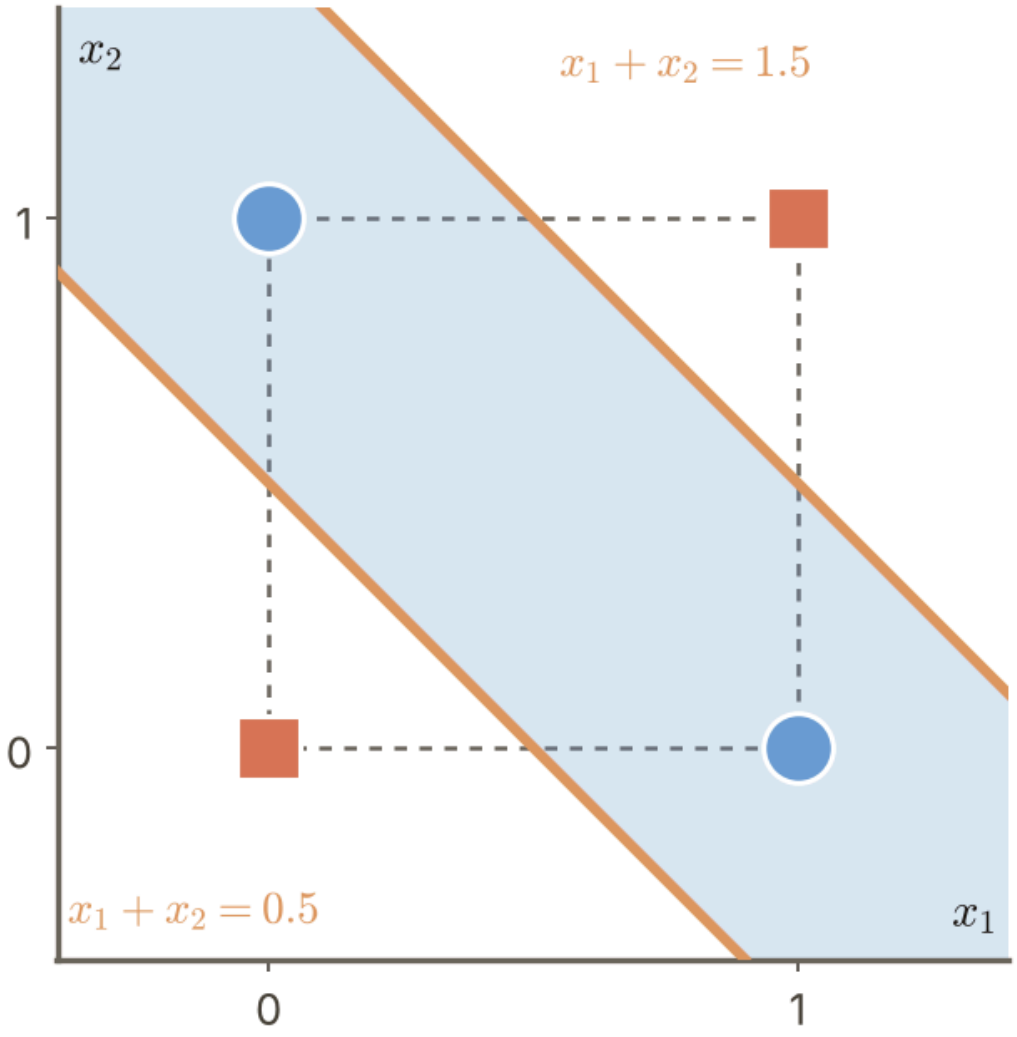


Figure `lec05-kp00-map`: top to bottom are KP01 to KP05.

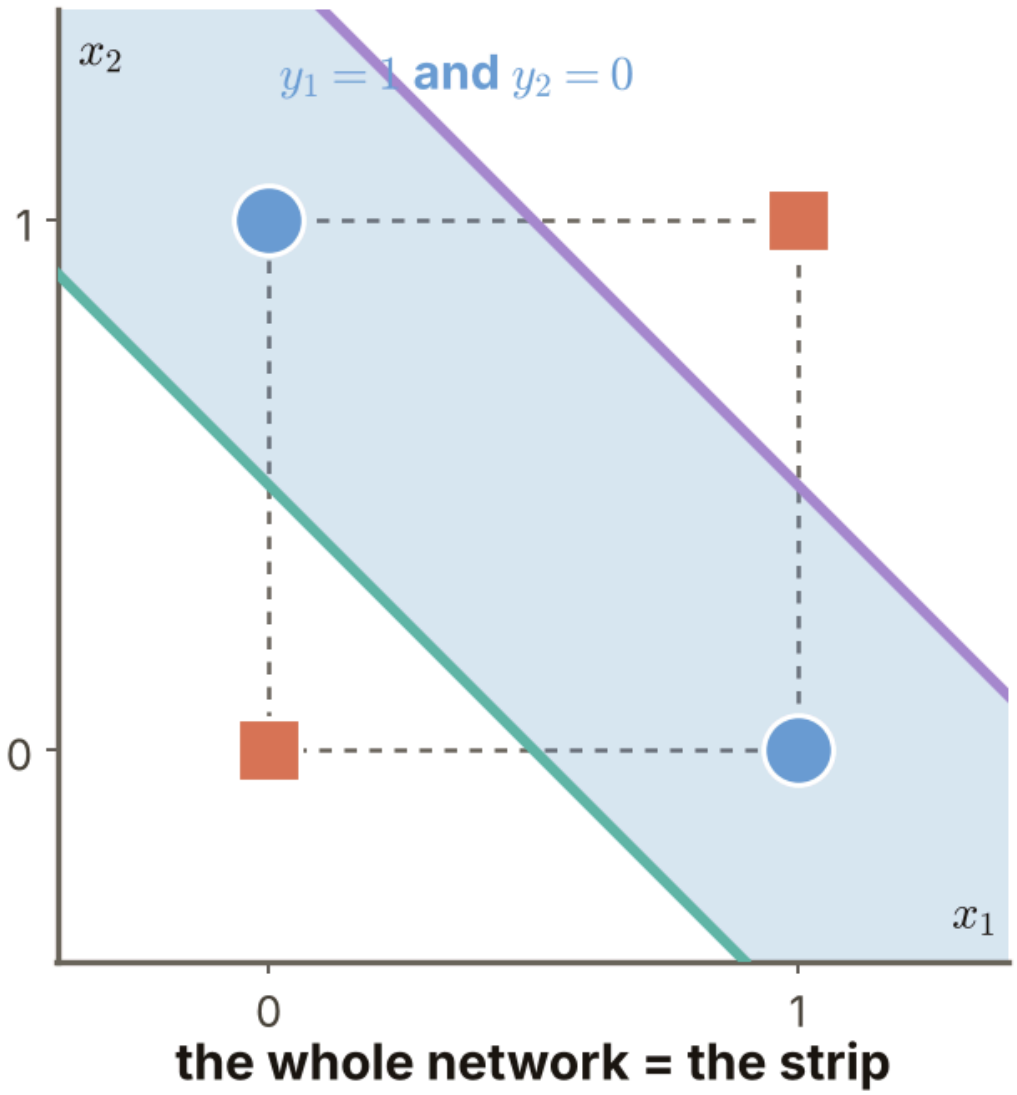
3. Question, answer and what is left open for the five knowledge points

Knowledge point	Question	Answer	Left open
KP01 one line fails	Does any line separate XOR?	The two A scores sum above 0 and the two B scores at most 0, yet it is the same expression, a contradiction.	What must change is the combination of boundaries, e.g. a strip between two lines.

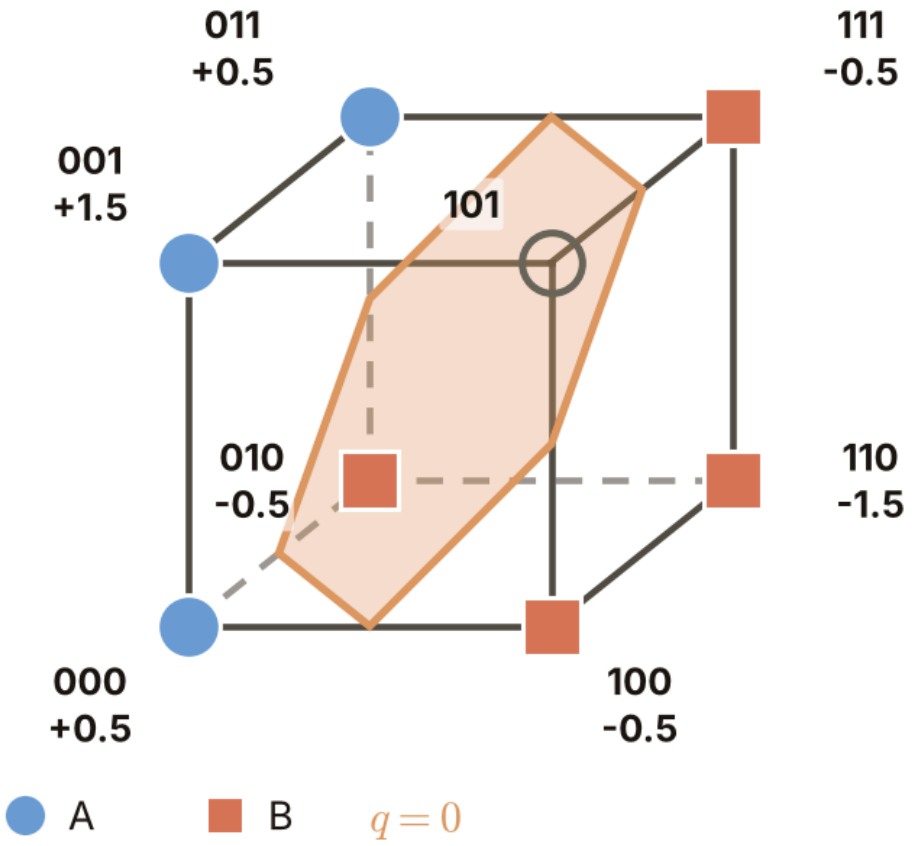
Knowledge point	Question	Answer	Left open
KP02 encode then classify	How are the two decisions built as a network?	Two lines give one bit each, the four points get three codes, one output line separates A from B.	With more lines, is the output layer still enough?
KP03 codes and limit	With more lines, can one cut separate every labelling?	Some yes, some no (A and B alternate on the bottom face).	Labellings one cut misses need another layer.
KP04 detect each code	How do we add a layer on top of the codes?	Detector score $= 0.5-$ (wrong bits), output layer merges with an OR.	Only proves it is enough; how does the network grow from points?
KP05 grow from points	With only labelled points, how does the network grow?	A master unit splits once, ancillary units handle mixed groups and add bits to purify.	All correct only means these points; fix the architecture and learn weights.

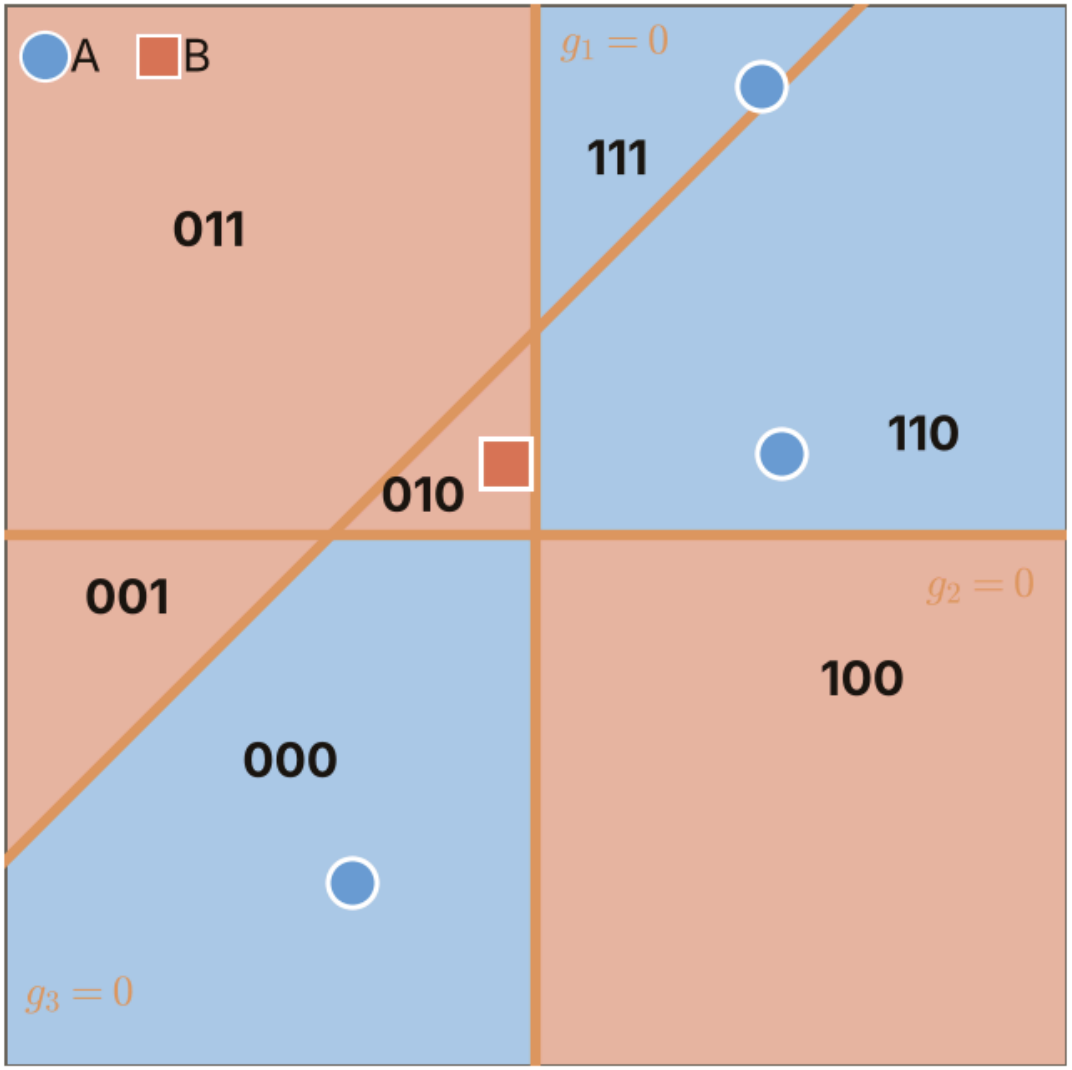


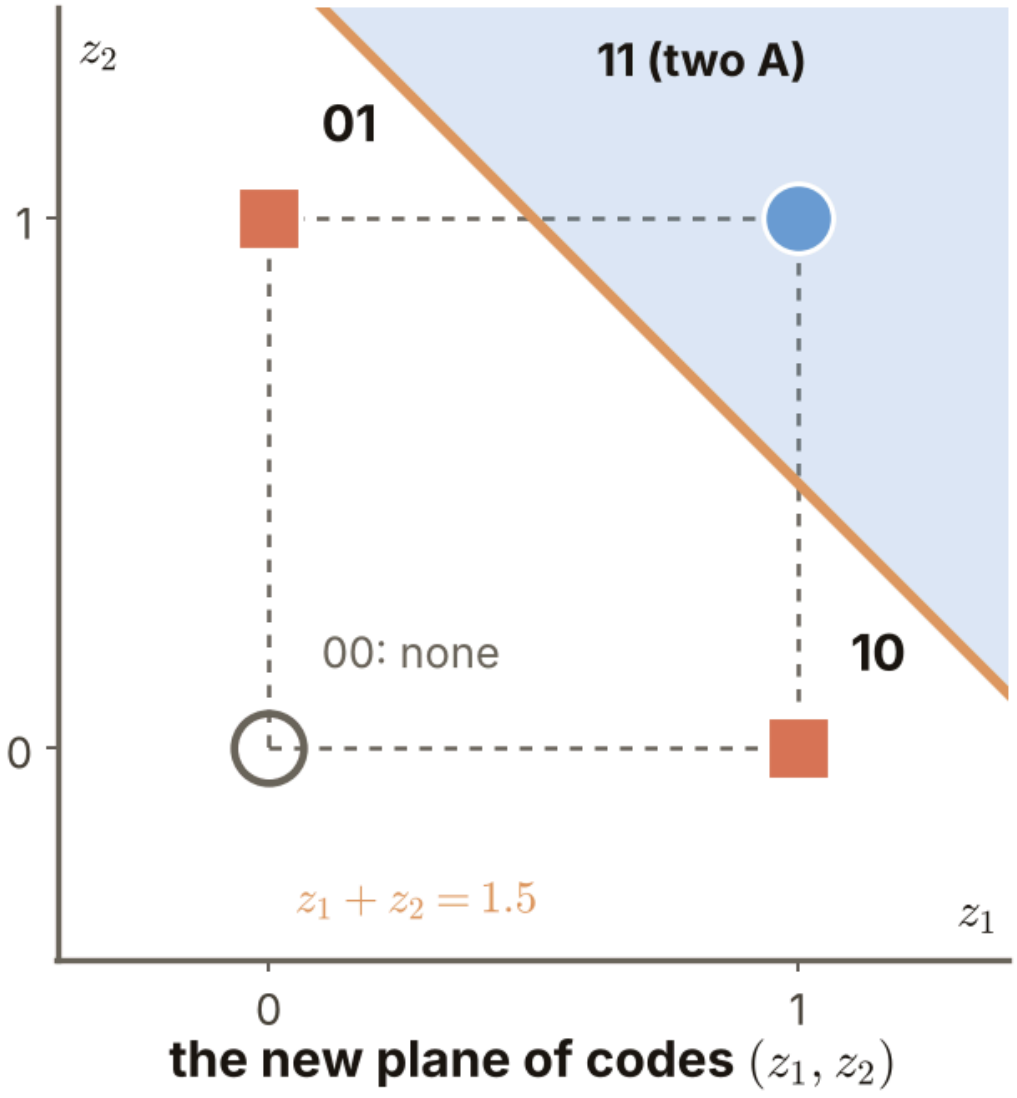
A lies between the two lines



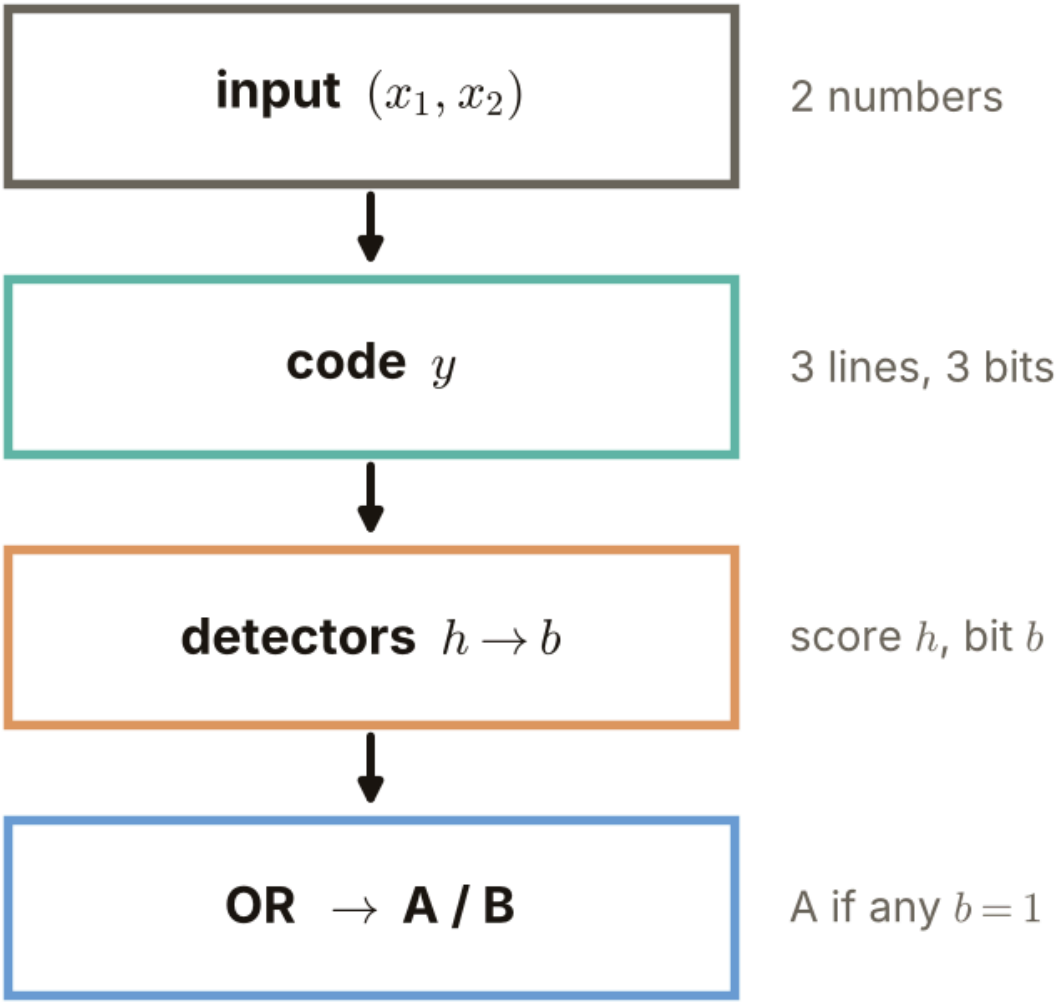
the whole network = the strip


011
+0.5
111
-0.5
001
+1.5
101
010
-0.5
110
-1.5
000
+0.5
100
-0.5
A
B
$q = 0$


A
B
$g_1 = 0$
111
011
110
010
001
$g_2 = 0$
100
000
$g_3 = 0$



4. Recap: the whole network and four reminders



Common misconception: that a construction showing a network is enough means the weights are learned or the network is minimal. A construction only proves it is enough.

Four reminders: one cut on fixed codes has a limit; contradictory labels cannot be all correct; all correct only means these points; a construction proves it is enough, not that the weights are learned or the network is minimal. Next chapter: fix the architecture first, then learn the weights with a cost function (backpropagation).

Terms and sources

English	Knowledge point
Bit, step rule, hidden layer, hidden code	KP02
Code, realized code, unrealized code, linearly separable	KP03

English	Knowledge point
Target code, detector, OR merge, three-layer network	KP04
Master unit, ancillary unit, mixed group, purification	KP05

Source: the slides and lectures of KP01–KP05 of this chapter (compiled 2026-10-25). This page only recaps and adds no new content.